

Ana Carolina Cunha

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EDUCATION

Technical University of Munich, Munich, Germany

April 2025 - Present

Master's of Science in Informatics, Specializing in Machine Learning and Analytics

University of California - San Diego, San Diego, CA

September 2019 - June 2022

Major: Bachelor of Science, Cognitive Science Specializing in Machine Learning and Neural Computation

Minor: Computer Science

EXPERIENCE

BMW, Munich, Germany

Software Engineer - Internship

May 2025 – Present

- Integrated sensor data used for lane boundary extraction into the internal fingerprint pipeline, supporting AI-based localization for autonomous vehicles using **Python**.
- Worked in a cloud-based environment using **AWS** services for data storage and processing.
- Authored onboarding and technical documentation, improving team efficiency and knowledge transfer.
- Collaborated with other engineers to translate data requirements into engineering solutions while adhering to confidentiality protocols.

Inscribe AI, Portugal

Software Engineer II

November 2024 – April 2025

- Doubled functionality for the AI compliance agent by adding new **NLP**-powered **RAG**-based compliance checks using **Python**, **Django**, and **AWS Bedrock**.
- Improved LLM retrieval accuracy by 7.5% by designing a **precision-recall balancing technique** to eliminate incorrect results while preserving relevant data.
- Aligned system capabilities with client requirements by participating in customer calls to discuss expectations for compliance analyst POCs.
- Part of the foundational team for the AI compliance analyst, contributing to the design and implementation of core features.
- Contributed to the frontend development of the AI compliance agent using **Vue.js**, ensuring seamless user experience and integration with backend services.

Attentive, New York, NY, USA

Software Engineer

August 2022 - September 2023

- Expedited operations within the Client Strategy team by 50% by building new client-facing features using **React**, **TypeScript**, **Java**, and **SQL**.
- Optimized the MOPS team's access to client data by 87.5% through upgrading internal tools with **Python**, **Airflow**, and automated data pipelines.
- Improved client data security by migrating 10+ services in **AWS** using **DynamoDB** and **SQL**, ensuring compliance with data protection protocols.
- Boosted email campaign feature adoption by showcasing real-time statistics through enhanced **React** and **TypeScript** components in the client UI and updated **GraphQL APIs**.
- On-call support for client emergencies, utilizing **DataDog** to debug, diagnose, and resolve complex backend issues.

Software Engineer - Internship

June 2021 - July 2022

- Revamped legacy client UI, modernizing it into a scalable web application and **REST API** using **React**, **TypeScript**, **Java**, and **SQL**, enhancing user experience.
- Simplified testing of end-to-end **webhooks** for clients by implementing new functionality in the backend using **Java** and integrating continuous testing automation.

Center of Global Justice at University of California - San Diego, San Diego, CA, USA

STEAM Education Intern

January 2020 – September 2021

- Provided technology education to 35+ children in Tijuana, Mexico by developing a computer literacy program.
- Enabled access to technology to 500+ community members in Tijuana through repurposing the university's e-waste.

PROJECTS

3D Face Reconstruction

Group Project for 3D Scanning and Motion Capture

Built a 3D face reconstruction pipeline using **Python** to recover facial geometry from a single 2D image using both BFM and FLAME morphable models. Implemented an energy-based optimization framework incorporating sparse landmark alignment, dense photometric consistency, and regularization terms to refine shape and expression parameters.

MedSearch

Personal Project

Developed a web-app using **React** and the **Perplexity API** that helps users research medically accurate information and generate structured reports. Applied prompt engineering techniques to optimize AI query accuracy and consistency.

AnaGPT

Personal Project

Developed a ChatGPT clone using **React**, **Node.js**, and the **OpenAI API** to simulate real-time conversational AI, integrating user authentication and state management.

Platformer Game

Personal Project

Leveraged the **Pygame** library in **Python** to build a platformer game.

SKILLS & CERTIFICATIONS

Languages: Python | JavaScript | TypeScript | Java | SQL | C++ | HTML | CSS

Tools & Frameworks: TensorFlow | PyTorch | Django | MySQL | PostgreSQL | Docker | React | Node.js | GraphQL | AWS | CI/CD Pipelines

LEADERSHIP

Women in CS, Munich, Germany

Member

May 2025 – Present

- Contribute to the development of events that support women pursuing an education in computer science.

Attentive Women Engineers, Attentive, New York, NY

Recruitment Board

January 2023 – September 2023

- Collaborated with the recruiting team to create events that increased women applicants in engineering, fostering diversity and inclusion within the company.

Girls Who Code, University of California - San Diego Chapter, San Diego, CA

Vice President of Web Development

June 2021 – June 2022

- Created a club website and workshops to teach web development, enhancing outreach and technical skills for members.

Association for Computing Machinery, University of California - San Diego Chapter, San Diego, CA

Team Mentor

September 2021 – January 2022

- Mentored a team in **Agile** development practices, guiding them through tools and instructional material to successfully complete their **React** project within the timeline.